



**Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.**

UNIT 7 • STANDARD 6.NOS.7

## Reflect Points Across Axes

Lesson 7-9 • Teacher Copy

**Name:** \_\_\_\_\_

**Date:** \_\_\_\_\_

**Class:** \_\_\_\_\_

## My Notes

Level 3 Enrichment



### TODAY'S OBJECTIVES

**Content Objective:** I can reflect points across the x-axis and y-axis on the coordinate plane.

**Language Objective:** I can explain my work using the words reflection, x-axis, y-axis, and symmetry.



See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.



Fill in each blank as we go. Use the Word Bank to help you.



### WORD BANK — FILL EACH BLANK WITH THE BEST WORD

Reflect Points Across Axes

Reflection

x-axis

y-axis

Symmetry

Integer

Perpendicular



Tap any word to see what it means and a picture.

Level 3: try the blanks from memory first, then check the bank.

1

— Flip a point across an axis so it stays the same perpendicular distance from that axis.

2

A flip of a point or shape over an axis to make a mirror image is a

.

3

The horizontal number line on the coordinate plane is the

.

4

The vertical number line on the coordinate plane is the

.

5 When two halves are mirror images of each other, the figure has

.

6 A whole number or its opposite, with no fraction part, is an

.

7 Two lines that meet at a  $90^\circ$  square corner are  .

## Write About the Math

Guided ESOL writing

Use the support level you need. Every level answers the same math question.

### 1. Understand the Question

EXPLAIN

**How do reflections help the game designer create a balanced, symmetric level?**

**Your job:** Answer the question. Use math evidence. Explain how the evidence proves your answer.

*Responde la pregunta. Usa evidencia matemática. Explica cómo la evidencia demuestra tu respuesta.*

## 2. Plan Your Math Words

Check at least two words you will use.

☐ **Reflect Points Across Axes** — Flip a point across an axis so it stays the same perpendicular distance from that axis.  
*Reflejar puntos sobre los ejes — Voltar un punto sobre un eje para que quede a la misma distancia perpendicular de ese eje.*

☐ **Reflection** — A flipped, mirror image across a line.  
*Reflexión — Una imagen reflejada al otro lado de una línea.*

☐ **x-axis** — The line that goes across the grid. Flipping over it changes the y sign.  
*Eje x — La línea que va de lado en la cuadrícula. Voltar sobre ella cambia el signo de y.*

☐ **y-axis** — The line that goes up the grid. Flipping over it changes the x sign.  
*Eje y — La línea que va hacia arriba en la cuadrícula. Voltar sobre ella cambia el signo de x.*

☐ **Symmetry** — When a shape looks the same on both sides of a line.  
*Simetría — Cuando una figura se ve igual a ambos lados de una línea.*

**Say it first:** Say your idea to a partner or quietly to yourself before you write.

*Di tu idea a un compañero o en voz baja antes de escribir.*

**The enemy reflected over the y-axis is at \_\_\_\_ because \_\_\_\_.** **The enemy reflected over the x-axis is at \_\_\_\_ because \_\_\_\_.** **Reflections help game designers because \_\_\_\_.**

*Di tu respuesta primero: Mi respuesta es \_\_\_\_ porque \_\_\_\_.*

### 3. Build Your Explanation

**Model the parts:** This model shows the parts of an explanation. It does not answer your writing question.

**Claim:** Reflecting over the y-axis changes the \_\_\_\_ coordinate because \_\_\_\_.

**Evidence:** Student states reflecting over the y-axis flips the sign of the x-coordinate while the y-coordinate stays the same, so (3,2) becomes (-3,2).

**Reasoning:** This evidence connects the definition of reflect points across axes to the mathematical claim.

#### Start

Most language support

Complete the frames. Use the word bank.

*Completa las oraciones. Usa el banco de palabras.*

- **The enemy reflected over the y-axis is at \_\_\_\_ because \_\_\_\_.** The enemy reflected over the x-axis is at \_\_\_\_ because \_\_\_\_.
- **My answer is \_\_\_\_.**

*Mi respuesta es \_\_\_\_.*

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#### Build

Some language support

Write two connected sentences. Use because, so, or but.

*Escribe dos oraciones conectadas. Usa porque, entonces o pero.*

- **My claim is \_\_\_\_.**
- **I know because \_\_\_\_.**

*Mi afirmación es \_\_\_\_.*

*Lo sé porque \_\_\_\_.*

- **So \_\_\_\_.**

*Entonces \_\_\_\_.*

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## Explain

### Light language support

Write a claim, give math evidence, and explain your reasoning.

*Escribe una afirmación, da evidencia matemática y explica tu razonamiento.*

- **My claim is \_\_\_\_.**  
*Mi afirmación es \_\_\_\_.*
- **My math evidence is \_\_\_\_.**  
*Mi evidencia matemática es \_\_\_\_.*
- **This proves \_\_\_\_ because \_\_\_\_.**  
*Esto demuestra \_\_\_\_ porque \_\_\_\_.*

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## 4. Check Your Explanation

- ☐ I answered the question.  
*Respondí la pregunta.*
- ☐ I used math evidence: numbers, an equation, a model, or a comparison.  
*Usé evidencia matemática: números, una ecuación, un modelo o una comparación.*
- ☐ I explained how the evidence proves my answer.  
*Explicué cómo la evidencia demuestra mi respuesta.*
- ☐ I used at least two math words.  
*Usé por lo menos dos palabras matemáticas.*
- ☐ I reread complete sentences and punctuation.  
*Volví a leer mis oraciones completas y la puntuación.*

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## Writing Guide — Teacher Copy

The support level changes the amount of language scaffolding, not the mathematical expectation. Look for the same five features in every response.

- I answered the question.
- I used math evidence: numbers, an equation, a model, or a comparison.
- I explained how the evidence proves my answer.
- I used at least two math words.
- I reread complete sentences and punctuation.



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## Answer Key & Teacher Guide

1. **Notes 1:** Reflect Points Across Axes
2. **Notes 2:** Reflection
3. **Notes 3:** x-axis
4. **Notes 4:** y-axis
5. **Notes 5:** Symmetry
6. **Notes 6:** Integer
7. **Notes 7:** Perpendicular
8. **Watch for this mistake:** *A common mistake in Reflect Points Across Axes is flipping the sign of the wrong coordinate — changing the x-coordinate when reflecting over the x-axis, instead of the y-coordinate (or vice versa for the y-axis). For example, to reflect (3, -5) over the x-axis, a student might flip the x-coordinate and write (-3, -5), but reflecting over the x-axis only changes the sign of the y-coordinate: (3, -5) becomes (3, 5). Before you submit, ask: 'Did I flip the sign of the coordinate that matches the axis I'm reflecting over — y for the x-axis, x for the y-axis?'*
9. **Try It 1:** A. (5, 3) — *Reflecting across the y-axis flips the sign of the x-coordinate only. (-5, 3) becomes (5, 3).*
10. **Try It 2:** A. The y-coordinate — *Across the x-axis the rule is  $(x, y) \rightarrow (x, -y)$ , so only the y-coordinate flips sign. (8, -6) becomes (8, 6).*